Section D — Student experiment (practical)

Title

Measuring the critical angle of glass and verifying n = 1/sin C

Aim

To measure the critical angle (C) for a transparent material, for example glass, and to confirm that 1/sin C is equal to the given refractive index, n, of the material. That is, to verify that **n = 1/sin C**.

Before the experiment — my prediction

The given refractive index of my block is n = _____

My predicted critical angle is $C = \sin^{-1}(1/n)$ = _____

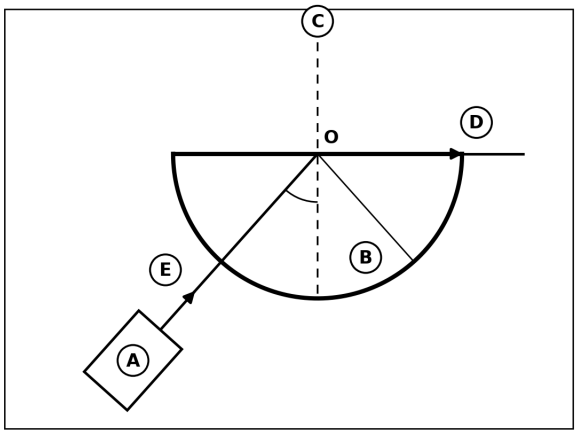
If the relationship holds, my measured value of 1/sin C should be close to _____

Apparatus

Ray box with a single slit · semicircular glass block · sheet of plain white paper · protractor · ruler · sharp pencil

Set-up

Identify each labelled part of the diagram. Choose from: ray box, semicircular glass block, normal, incident ray, emergent ray along the flat face.



A _____

B _____

C _____

D _____

E _____

Procedure

1. Place the semicircular block, flat side down, on the paper. Draw around it and mark the centre of the flat face as O.
2. Remove the block. Draw the normal at O, perpendicular to the flat face. Replace the block exactly in its outline.

3. Aim a narrow ray from the ray box at the curved surface so that it travels towards O. Move the ray box around the curved edge, keeping the ray pointed at O.
4. Slowly increase the angle of incidence at the flat face until the emergent ray just disappears along the flat face. Decrease it slightly until it just reappears. The critical angle lies between these two positions.
5. Mark two crosses on the incident ray, at least 5 cm apart. Mark the point where the emergent ray lies along the flat face. Remove the block.
6. Draw the incident ray through your two crosses to O. Measure the angle between the incident ray and the normal. This is C.
7. Repeat for five trials, starting again from the ray box each time. Complete the data table.

Why does the ray not bend as it enters the curved surface? _____

_____

Why must the ray always point at O? _____

_____

Data table

Trial	Critical angle C (°)	sin C	n = 1 / sin C
1			
2			
3			
4			
5			
Mean			

Analysis

Given n = _____ Experimental n (from the mean) = _____

Percentage difference = (experimental − given) ÷ given × 100 = _____

Spread in C = (largest − smallest) ÷ 2 = _____ °

Lowest n = 1 / sin (C + spread) = _____

Highest n = 1 / sin (C − spread) = _____

Does the given value of n lie inside this range? _____

Precautions — say why each one improves the result

Precaution	Why it improves the result
Use a narrow ray	
Mark crosses far apart on the ray	
Look straight down when marking points	
Repeat five times and take the mean	

Sources of error

Name two sources of error. For each one, say whether it is likely to make C too large or too small.

_____

_____

_____

Secondary data — check the model against published values

The table gives typical published values from reference tables. Calculate sin C and 1/sin C for each material and decide whether the published data support n = 1/sin C.

Material	Published n	Published C (°)	sin C	1 / sin C	Agrees?
Water	1.33	48.8			
Perspex	1.49	42.2			
Crown glass	1.52	41.1			
Diamond	2.42	24.4			

PEER conclusion — complete after the experiment

P	**Point** — what did you find? State whether your data verify n = 1/sin C.
E	**Evidence** — what data supports it? Quote your mean value of n, the given n, and the percentage difference.
E	**Explanation** — why does this happen physically? Name Snell's law and the reversibility of light in your answer.
R	**Reflection** — what would you improve and why? Refer to the secondary data.